

MR. KEVIN WOODS, EIT

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EDUCATION

VILLANOVA UNIVERSITY, Center for Engineering Education and Research

Villanova, PA

Doctor of Philosophy in Mechanical Engineering

May 2012

- Thermo-Fluid Systems Research

Master of Science in Mechanical Engineering

May 2008

- **Thermo-Fluid Systems Graduate Certificate – May 2008**

- Recipient of Villanova BS/MS Graduate School Scholarship

- Overall Graduate GPA: 3.70/4.00

- Programming: Fluent, Matlab, MathCAD, AutoCAD, Gambit, Labview, and Microsoft Office

- Machine Experience: Scanning electron microscope, Mercury Spectrometer, General Thermal Laboratory Equipment, General Chemistry Laboratory Equipment

Bachelor of Science in Mechanical Engineering

May 2007

- Magna Cum Laude, Overall Undergraduate GPA: 3.75/4.00

- **Passed Fundamentals of Engineering Exam – April 2007**

- Member of Pi Tau Sigma: Mechanical Engineering Honor Society

- Member of Tau Beta Pi: Engineering Honor Society

- Member of Phi Kappa Phi: All Academic Honor Society

- Villanova Award for Academic Excellence

- Dean's List all semesters

WORK AND RESEARCH EXPERIENCE

LABORATORY FOR ADVANCED THERMAL AND FLUID SYSTEMS

Villanova, PA

Graduate Research Assistant, Dr. Alfonso Ortega

Jan. 2007 – Present

- **American Refining and Biochemical Geothermal Research Project:** Designed and built a well-bore simulator for water well intake simulation for geothermal heating, ventilating and air conditioning systems. Gained valuable technical knowledge on project planning, geothermal technology and electrical systems.
- **The Benjamin Franklin Institute Medical Device Research Project:** Modeled, designed and built multiple prototype medical thermal management devices for treatment of hot flashes using thermo-electric coolers.
- **ASME 2007 Sustainability Conference in Long Beach, CA**
- **ASME 2008 R&D Symposium in Washington D.C.:** Lobbied for an increase in research funding for the NSF and DOE for fiscal years 2008 and 2009.

NAVAL RESEARCH ENTERPRISE INTERN PROGRAM (NREIP)

Philadelphia, PA

Carderock Division, Philadelphia Naval Yard

June 2006 – August 2006

- **Unmanned Surface Vehicle Research:** Designed and began implementation of a hybrid test bed for an unmanned surface vehicle.

VILLANOVA UNIVERSITY TEACHING ASSISTANT/INSTRUCTOR

Villanova, PA

Mechanical Engineering Department

Jan 2008 – May 2008

- **Thermodynamics I Teaching Assistant for Dr. Alfonso Ortega:** Main tasks include teaching a practicum once a week, grading homework and exams, and teaching class for Dr. Ortega when away at conferences.

VILLANOVA UNIVERSITY LABORATORY INSTRUCTOR

Villanova, PA

Mechanical Engineering Department

Aug 2007 – Dec 2008

- **Flat Plate Heat Exchanger and Double Pipe Heat Exchanger Lab Instructor:** Instructed Senior Mechanical Engineering Students in a laboratory course. Performed analyses on heat transfer from a heated flat plate, and heat exchanger effectiveness from a shell and tube heat exchanger.

VILLANOVA UNIVERSITY RESIDENT ASSISTANT
Residence Life

Villanova, PA
Aug 2007 – May. 2008

- In charge of 36 residents in St. Marys Residence Hall.

ENVIRONMENTAL LABORATORY, Villanova Center for the Environment

Villanova, PA

Undergraduate Research Assistant, Dr. Rominder Suri

Jul. 2004 – May 2006, Aug. 2006 – Nov. 2006

- **Pharmaceutical Wastewater Research Project:** Analyzed the effect of granular activated carbon on pharmaceutical wastewater containing thimerosal, an organic mercury compound used in vaccines. Co-authored a published paper in the Journal of Hazardous Materials.
- **Ultrasonic Improvement of Acid Neutralization Research Project:** Analyzed the effect of ultrasonic waves on the reactivity of lime for acid neutralization.
- **United States Geological Survey (USGS) Survey Project:** Surveyed hormonal contaminants in streams and rivers in Chester County.
- **International Conference for Energy, Environment and Disasters in Charlotte, NC in 2005**

PUBLICATIONS

Pharmaceutical Wastewater Research Project:

- Velicu, M., Fu, H., Suri, R.P.S., Woods, K., (2007) "Use of adsorption process to remove organic mercury thimerosal from industrial process wastewater" Journal of Hazardous Materials, 148(2007) 599–605.

LANGUAGES/SKILLS/INTERESTS

Clubs: -American Society of Naval Engineers: Held Title of Secretary for 2006-2007 Academic Year.
 -American Society of Mechanical Engineers
Languages: -Lived in Mexico City for 2 years and Tokyo for 7 years, fluent in Spanish and proficient in Japanese.
Volunteer: -Water for Waslala service trip to Nicaragua for water system consultation and design.
Other: -Completed Fundamentals of Engineering Exam in April 2007